Power & Protection

文件: 01_power.kicad_sch

MCU & Debug

文件: 02_mcu.kicad_sch

IMU

文件: 03_imu.kicad_sch

TTL Bus AUTO

Feetech TTL slave ID 200 + trunk power hub for MicroDuck HD-1910 build

**IMU-to-TTL**

Sheet: /
File: imu_to_ttl.kicad_sch

***Title: IMU-to-TTL board***

| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 1/6 |

**POWER & PROTECTION — battery in, trunk power hub**

J1 wire side: keyed XT30 on battery cable. Measure polarity before first plug.

Q1: D=VBAT_FUSED S=VBAT G via 100k/1M + 100nF -> reverse protection with ~10ms soft start.

Per-branch current budget: PH 2.0 contact = 2A. J_NECK also feeds the head HAT.

#FLG21 PWR_FLAG
#FLG22 PWR_FLAG
#FLG23 PWR_FLAG
#FLG24 PWR_FLAG
#FLG25 PWR_FLAG

VBAT_RAW
VBAT_FUSED
VBAT
VBAT_LOGIC
GND

C2 100nF 50V (VBAT, GND)
C4 100nF 50V (VBAT_LOGIC, GND)
C6 100nF 5 (partial)
D1 SMBJ12A (VBAT, GND)
C1 10uF 25V X7R (VBAT, GND)
D2 SMF12A (VBAT_LOGIC, GND)
C3 10uF 25V X7R (VBAT_LOGIC, GND)
C5 10uF 1 (partial)

J1 BATT IN 2.54 2P (VBAT_RAW, GND; B+, B-)

F1 FUSE 5A 2512 (VBAT_RAW, VBAT_FUSED)

Q1 AO4407C (VBAT x3, Q1_GATE, VBAT_FUSED x5)

F2 PPTC 200mA (VBAT, VBAT_LOGIC)

U4 HT7533-1 (VBAT_LOGIC, VIN, VOUT, +3V3, GND)

R1 1k (partial)

C14 100nF 50V (Q1_GATE, GND)

R9 100k (Q1_GATE, GND)

R10 1M (Q1_GATE, VBAT)

TP8 HAT_PWR+ DNP (VBAT)

TP9 HAT_PWR- DNP (GND)

Feetech TTL slave ID 200 + trunk power hub
**IMU-to-TTL**

Sheet: /Power & Protection/
File: 01_power.kicad_sch

***Title: Power & Protection***

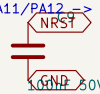
| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 2/6 |

**MCU — STM32G031F8P6 (TSSOP-20)**

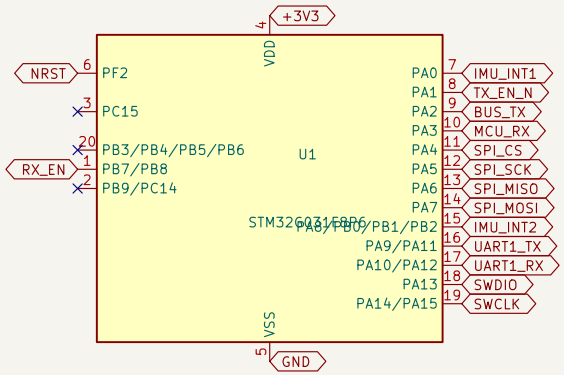
Clock: HSI16 only. No crystal — PC14 is LSE, NOT HSE (initial-review fix).

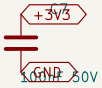
USART1 debug on pins 16/17 needs SYSCFG remap (PA11/PA12 -> PA9/PA10).

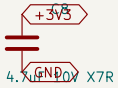


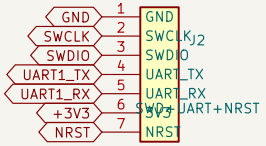




Unused pins: default analog high-Z after reset; left unconnected by design.




Feetech TTL slave ID 200 + trunk power hub

**IMU-to-TTL**

Sheet: /MCU & Debug/
File: 02_mcu.kicad_sch

***Title: MCU & Debug***

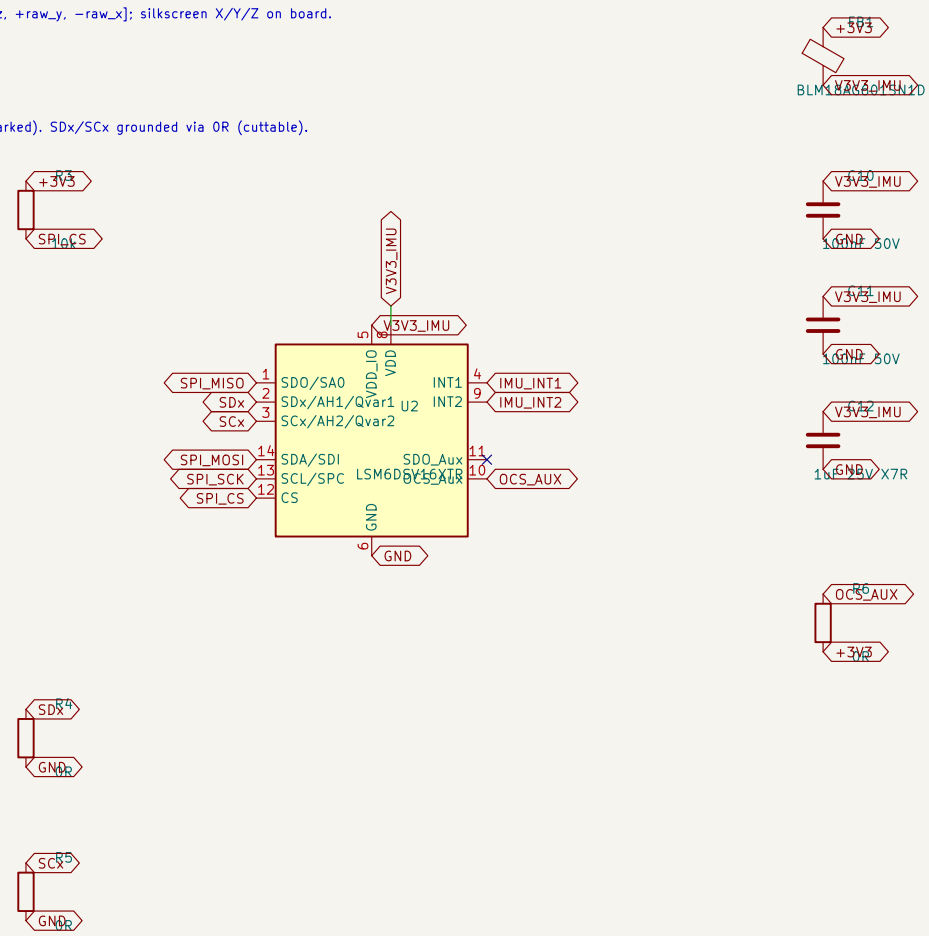
| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/6 |

**IMU — LSM6DSV16X (SPI + SFLP)**

Gyro full scale must be +/−500dps to match duck-control decode.

Mount axis: trunk = [+raw_z, +raw_y, −raw_x]: silkscreen X/Y/Z on board.

SDO_Aux unused -> NC (marked). SDx/SCx grounded via 0R (cuttable).



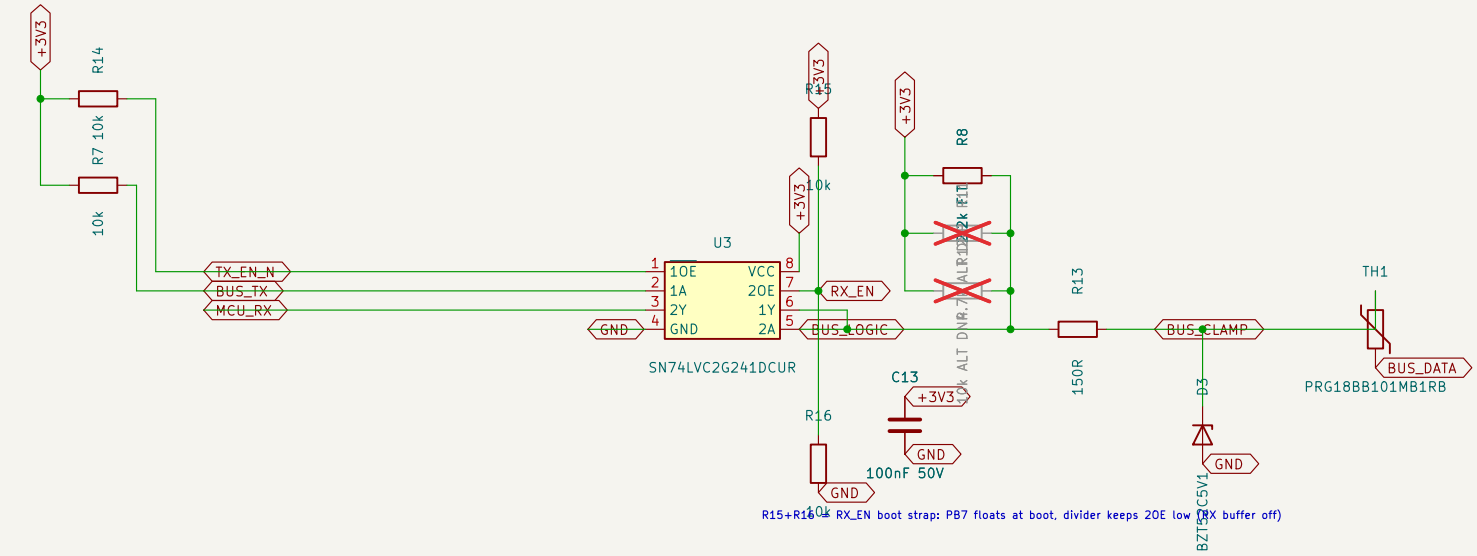

Feetech TTL slave ID 200 + trunk power hub

**IMU-to-TTL**

Sheet: /IMU/
File: 03_imu.kicad_sch

***Title: IMU***

| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/6 |




**TTL BUS v2 — 2G241: PA1=TX_EN_N(1OE#) + PB7=RX_EN(2OE) + R16 boot strap (matches real board)**

PA3 RX is direct on BUS_LOGIC: FT 5V-tolerant pin + 150R + 5.1V zener + 100R PTC.

Echo: during TX the UART sees its own bytes on RX — firmware must ignore RX while transmitting.

1Mbps timing gate: SPICE sweep (2.2k/4.7k/10k x 100/200/300pF) before PCB.

R8/R11/R12 = BUS_LOGIC pull-up options, fit ONE (2.2k default / 4.7k / 10k for trunk length)

Feetech TTL slave ID 200 + trunk power hub

**IMU-to-TTL**

Sheet: /TTL Bus AUTO/

File: 04_bus.kicad_sch

***Title: TTL Bus AUTO***

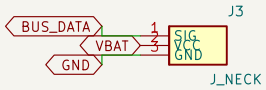
| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 5/6 |

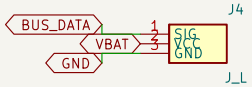
CONNECTORS — Feetech PH 2.0, all same pinout

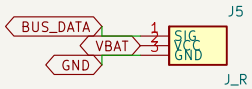
PINOUT: 1=SIG(TTL) 2=VCC(bus) 3=GND — HD-1910 spec A/0. NOT Dynamixel order!

All three bus ports are one shared net; VDD/GND pass through with wide 2oz copper.

Per-port current <= 2A (PH contact). J_NECK also carries the head HAT input.



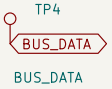






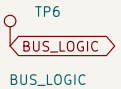
















Feetech TTL slave ID 200 + trunk power hub

**IMU-to-TTL**

Sheet: /Connectors & Test/
File: 05_connectors.kicad_sch

***Title: Connectors & Test***

| Size: A4 | Date: 2026-09-16 | **Rev: v1.0** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 6/6 |